



Antifouling and anti-algal effects of chitosan nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films on marine microalgae *Dunaliella salina*

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ABSTRACT

Biofouling is the main problem for marine water systems such as the immersed leisure vessels, oceanographic sensors, heat exchangers, and ship hulls. The obstacle of biofouling created by barnacles, mussels, bacteria, and micro and/or macro algal species. To overcome this problem, some metal nanoparticles incorporated films and membranes were employed to control the biofouling. In this study, metal nanoparticles (Ag NPs and TiO₂) of different concentrations (0.25, 0.5, 0.75 wt %) were incorporated with chitosan polymer to fabricate nanocomposite and pristine films. The biofouling inhibition was examined using the marine microalgal species *Dunaliella salina*. The higher performance in algal viability ($45.42 \pm 0.36\%$) was observed by TiO₂ and concentration maintained Ag NPs incorporated chitosan nanocomposite (0.5 wt% Ag/ 0.75 wt% TiO₂) films under UV-C condition, in comparison with other nanocomposite and pristine concentrations. The same concentration showed higher activity in the slime formation (Glycocalyx test), mass assessment, exo-polymeric substances (EPS), and membrane damages (LPO) under UV-C condition. The release of silver ions (Ag⁺) and uptake of Ag showed good performance of Ag NPs incorporated nanocomposite film at higher concentration (0.5 wt% TiO₂/ 0.75 wt% Ag). The surface morphology of nanocomposite were characterized using SEM, and AFM, and the surface modifications of chitosan nanocomposites were examined by XRD, and FTIR. TiO₂ and Ag NPs being photoactive induce more ROS that damage the cell membrane leading to algal cell death and control of biofouling formation.

1. Introduction

Biofouling, in particular marine biofouling, is one of the major concerns in the present day scenario. It is a natural phenomenon, wherein organic matter and biofilms get accumulated and eventually helps in the attachment of alga, mussels, barnacles etc. [1]. This can create major issues and has economic effects on the nautical ecosystems since manmade submerged surfaces under seawater like leisure vessels, oceanographic sensors, heat exchangers, and ship hulls are highly susceptible to biofouling [2]. Marine transportation systems usually face problems concerning unnecessary liquid gas consumption because of the increased hydrodynamic mass and friction haul, thus decreasing the speed and manoeuvrability [3]. Generally, different kinds of antifouling paints are used to prevent the biofouling of the structures. However, the presence of chemicals like tributyltin and triphenylboranepyrindine (TPBP) in those paints indicated biocidal effects, and

hence, they were banned by the environmental organizations for use as antifouling agents in aquatic environments [4,5].

Preparation of antifouling membranes uses nanoparticles (NPs) for two major reasons. One of the reasons is the capacity of bonding between polymer chains and/or solvents and NP surface, which can lead to the fabrication of nanoparticle-incorporated polymer membranes. These modified structures play vital roles in ultra and nanofiltration membranes that can aid in separation with high selectivity and permeability for gas. The second reason is the tunability of nanoparticles, by which they can have hydrophilic nature and functional groups, thus, minimizing membrane fouling caused by aquatic organisms [6–9].

Metal NP-incorporated membranes play an important role in marine ecosystem due to their relative nontoxic nature [10]. Recent reports suggested the blending of metal nanoparticles such as Al₂O₃, ZnO, SiO₂, Fe₃O₄, TiSO₄, ZrO₂, TiO₂, Ag, attapulgite, zeolite, and carbon nanotubes with polymers to provide antifouling activity [11–20]. Among these

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NPs, TiO₂ NPs have good photocatalytic activity, hydrophilicity, and are cost effective [21]. The photocatalytic property of TiO₂ NPs makes them one of the best antimicrobial materials [22]. The usage of supporting NPs such as Au, Sn, Pt, and Ag can modify the TiO₂ catalysts and enhance the photocatalytic activity by limiting the electron-hole recombination [23]. Silver (Ag) NPs are known to show concentration-dependent anti-algal activity [24]. UV irradiation is another major part of biofouling control. Among all three UV radiations, UV-C irradiation is the most harmful and can elicit biocidal effects [10].

Blended polymer materials are used worldwide because of their three significant antifouling characteristics, which are prevention of coatings from organisms, releasing of existing coatings, and inhibition of the organisms by degrading their coatings [3,25,26]. Chitosan being biocompatible and convenient for antifouling applications, is considered as one of the best TiO₂ and Ag nanoparticle-incorporating polymer. Using green chemistry processes, seafood by-product can be used to extract chitosan [27–30]. Recently, the researchers are focusing on modifying the chitosan to improve the polymer roughness, permeability, tensile strength, and fouling barrier by incorporating, blending, and grafting different nanomaterials [31–33]. Because of its backbone with charged groups (cationic properties) and ionic bonding with the algal outer cell membrane, chitosan by itself shows good antimicrobial properties [34]. Chitosan nanocomposites with incorporated NPs exhibited enhanced antimicrobial effects [35]. Deacetylation of chitin in chitosan improves the anti-algal effects of metal oxide nanocomposites, and such nanocomposites have been used as a coating material to make effective antimicrobial agents [36–38]. Recent study by Hosseini et al. demonstrated that the antimicrobial activity of chitosan depends upon the environmental factors such as type of test microorganisms, pH, and neighbouring compounds. Factors relating to chitosan such as deacetylation degree, derivative form, molecular mass, and concentration also play a role in determining the toxicity level in microorganisms [39].

Anti-algal and antifouling activity of TiO₂ and Ag NPs incorporated in a polymer matrix on *Dunaliella* sp. and other algal species have been reported recently. Ag/graphene (GAg) composite material inhibited 70% growth rate in *Dunaliella* sp. after 96-h interaction [40]. A comparative study between the effect of Ag NPs and Ag ions towards *Dunaliella salina* was performed by Sendra et al. and they observed an EC₅₀ value of 0.64 mg/L [41]. Another study revealed that silver-zeolite clusters inhibited 81% of biofilm formation and cell growth in *Dunaliella* sp. [42]. Yee et al. showed that PMMA/Ag NPs/GO nanocomposite could influence the antifouling activity against *Halomonas* sp. [43]. The same authors analysed the performance of Ag/TiO₂ nanocomposite on marine organisms, and these nanocomposites indicated better inhibition of the contaminants [43]. Jalali et al. synthesised Ag/TiO₂ incorporated different silane (3-mercaptopropyl trimethoxysilane and tetraethylortho silicate) sol-gel matrix and investigated the antibacterial activity of those matrix against *E. coli*, *S. typhimurium*, *S. aureus*, and *B. cereus* sp. [44]. Studies with other nanocomposites were also reported. Tabatabaei et al. prepared the bionanocomposite, SiO₂ NPs reinforced gelatin-k-carrageenan films and concluded that the improved physical properties of the SiO₂ reinforced bionanocomposite gelatin-k-carrageenan films were not competing with other synthetic and/or biocomposites (chitosan) films [45]. Eduok et al. reviewed the growth inhibition of *Dunaliella tertiolecta* by PDMS (polydimethylsiloxane) coatings and found that the PDMS coatings enhanced the fouling inhibition behaviour against the foulants [46].

In our present study, the anti-algal and antifouling activities of TiO₂- and Ag-incorporated nanocomposites and pristine chitosan films (chitosan having either TiO₂ or Ag NPs) were examined on marine algal species, *Dunaliella salina*, kept under UV–C and dark conditions. As mentioned above, it has already been established that TiO₂- and Ag-incorporated nanocomposites showed antifouling effects [43], and other groups of researchers have shown that UV radiations have biocidal effects, which also provide antifouling effects [10]. Hence, to the

best of our knowledge, none of the works involved studying the combinatorial effect of nanocomposites under UV–C irradiation on marine water algae, *Dunaliella* sp. TiO₂ NPs being photo-catalytically active, will produce a large amount of ROS under UV exposure. It was hypothesized that under UV–C radiation, ROS generated from TiO₂ NPs would in turn damage the algal cells and prevents the slime formation. Thus, in the current study, the potential antifouling ability of TiO₂/Ag nanocomposite under UV–C exposure was explored. Their cytotoxicity towards algal cells was evaluated by anti-algal activity, LPO estimation, and EPS quantification. The antifouling property of nanocomposite was checked using glycocalyx formation and mass assessment. Fourier-transform infrared spectroscopy (FTIR), atomic force microscopy (AFM), X-ray powder diffraction (XRD), and scanning electron microscopic (SEM) analyses were performed to characterize the treated and untreated nanocomposite and pristine films.

2. Materials and methods

2.1. Chemicals

The materials used for the formation of films i.e. chitosan, TiO₂ NPs [anatase (18 nm); CAS No: 1317-70-0], Ag NPs (14 nm) (CAS no: 7440-22-4) [TEM images of TiO₂ (Figure S⁻¹A) and Ag NPs (Figure S⁻¹B) were provided in the supplementary index], and the testing reagents (phenol and sulphuric acid) were acquired from Sigma-Aldrich. Chemicals used for assessments viz., toluidine blue, tetraboric acid, trichloroacetic acid, vitamin B12 (cyanocobalamin), and B1 (thiamin) were procured from Hi-media Laboratories Pvt. Ltd (Mumbai, India). Chloroform, used as a test solvent, was bought from SRL.

2.2. Nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) film formation

Chitosan (3 wt%) nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films were prepared according to the protocol mentioned in our previous paper [47]. The process is mentioned here in brief. The reason for the selection of 3 wt% chitosan is that it completely got dissolved at 65 °C compared to other concentration of chitosan. TiO₂ and Ag NP suspensions were prepared using a high-speed probe ultra-sonicator (Oscar ultrasonic cleaner, Microclean-109). 30 mg of chitosan powder was mixed with 30 mL of acetic acid, and then, this solution was added to different concentrations (0.1, 0.5, and 0.75 wt%) of homogenized TiO₂ and Ag NPs. These colloidal mixtures were then poured into fresh petri plates and allowed to air-dry at 40 °C for 24 h. The dried nanocomposite and nano pristine films thus formed were stripped off from the petri plate and stored at room temperature in a desiccator for further use (TiO₂ and Ag concentrations in the films are provided in Table S1, Supplementary information).

2.3. Test species

The marine algal species of *Dunaliella salina* was obtained from Central Marine Fishery Research Institute (CMFRI), Rameshwaram, Tamilnadu, India. The sub-cultured algal species were maintained in Conway medium (Table S2, Supplementary information) by providing a day/night rhythm of 16/8 h using a white fluorescent (TL-D super 80 linear fluorescent tube, Philips, India) light with 3000 lx intensity.

2.4. Estimation of toxicity

Dunaliella sp. cells were obtained during their exponential growth (21 days) stage by centrifuging the culture at 7000 rpm for 10 min, and the pellet was isolated. Using artificial sea water (ASW) (Component in Table S3, Supplementary information), the pellet was diluted to get an optical density (OD) of 0.1 (which contains 1×10^5 cells). This adjustment of OD was done based on the guidelines of OECD [48]. These diluted algal cells (10 mL) were then interacted with pure chitosan,

nanocomposite, and pristine films ($1\text{ cm}^2 \times 1\text{ cm}^2$) for 72 h under two different conditions: UV-C (TUV 15 W/G15 T8, Philips, India; 0.44 mW cm^{-2} intensity, wavelength $< 280\text{ nm}$) and dark (by covering the samples with an opaque sheet). After 72-h interaction with different concentrations of nanocomposite (TiO_2/Ag) and pristine (TiO_2 and Ag) films, the algal cell viability was estimated using an MTT assay [49]. After the interaction period, $20\text{ }\mu\text{L}$ of MTT was added to $500\text{ }\mu\text{L}$ of algal cells and was incubated for 4 h in dark. After incubation, the pellet was separated by centrifuging the samples at 8000 rpm for 8 min, and the obtained pellet was washed with ASW. Then, the pellet was dissolved in Dimethylsulphoxide (DMSO) to form the purple colour formazan. The absorbance of the formed formazan was measured at 570 nm using a microtiter plate reader. The viability of the control cells was taken as 100%, and the viability of the treated cell samples was determined with respect to the control.

2.5. Evaluation of lipid peroxidation

The lipid peroxidase (LPO) released from *Dunaliella* sp. interacted with nanocomposite (TiO_2/Ag), pristine (TiO_2 and Ag), and pure chitosan films under UV-C and dark conditions was determined indirectly by measuring the malondialdehyde (MDA) produced using the method of Piotrowska-Niczyporuk et al. [50]. After exposure, the control and treated samples were centrifuged at 7000 rpm for 10 min, and the pellets obtained were resuspended using 2 mL of TBA 0.25% (w/v) prepared in 10% TCA (w/v). The mixtures were then heated at $95\text{ }^\circ\text{C}$ for 30 min and were cooled down at room temperature. Further, they were again centrifuged at 7000 rpm for 10 min, and the absorbance of the supernatant was measured at 532 nm and 600 nm using an ELISA reader (Biotek, Power wave XS2, U.S.A). Unspecific turbidity was cancelled out by subtracting the absorbance observed at 600 nm from that at 532 nm .

2.6. Quantification of extracellular polysaccharides (EPS)

The release of extracellular polysaccharide (EPS) from algal cells after interaction with pure chitosan, nanocomposite (TiO_2/Ag), and pristine (TiO_2 and Ag) films was estimated by phenol-sulfuric acid method [51]. The interacted and control algal cells were centrifuged for 10 min at $10,000\text{ g}$, and the harvested pellet was resuspended in pre-chilled ethanol and incubated at $4\text{ }^\circ\text{C}$ for overnight to precipitate the carbohydrate. After overnight incubation, the precipitates were collected by centrifugation at $10,000\text{ g}$ for 10 min. To the precipitated carbohydrates, 2 mL of distilled water was added. From this, $50\text{ }\mu\text{L}$ of carbohydrate sample was taken and mixed with $150\text{ }\mu\text{L}$ of conc. H_2SO_4 , and to this, $30\text{ }\mu\text{L}$ of 5% phenol was added immediately. The mixture was then heated at $90\text{ }^\circ\text{C}$ for 5 min in a static condition. Well plates were then allowed to cool down, and the absorbance was analysed at 490 nm using a microtiter plate reader [52].

2.7. Glycocalyx formation and mass assessment

The inhibition of glycocalyx formation in *Dunaliella* sp. interacted with nanocomposite (TiO_2/Ag) and pristine (TiO_2 and Ag) films was evaluated using a series of fixation followed by staining steps.

Before staining, the treated films were washed twice with 1 mL of Milli-Q water and interacted with Carnoy's solution [mixture containing chloroform, ethanol, and glacial acetic acid in the ratio 3:6:1] for 30 min. Post-interaction, the films were stained with 0.1% toluidine blue for 1 h. The excess dye was washed off with 3 mL of milli-Q water. Then, 0.2 M of NaOH was added, and the films were heated for 1 h at $85\text{ }^\circ\text{C}$. The solution obtained was cool down at room temperature, and its absorbance was measured using a UV-vis spectrometer at 590 nm [53].

Before and after interaction of pure chitosan, nanocomposite (TiO_2/Ag), and pristine (TiO_2 and Ag) films with algal cells (*D. salina*), their

mass was assessed using the protocol by Chapman et al. [53]. After interaction, the films were washed with Milli-Q water and allowed to dry until it reaches a constant weight. The average mass of adhered algae on the pure chitosan, nanocomposite (TiO_2 and Ag), and pristine films (TiO_2/Ag) was then computed.

2.8. Silver ion release

The quantification of Ag^+ released from nanocomposite and pristine films was estimated by atomic absorption spectroscopy (AAS). Nanocomposite (TiO_2/Ag) and pristine (TiO_2 and Ag) films of a test size ($1\text{ cm}^2 \times 1\text{ cm}^2$) were cut and immersed in 10 mL of ASW under abiotic condition (without algae) for a period of 72 h. Post-interaction, the sample suspension was filtered with a $0.1\text{-}\mu\text{m}$ filter, followed by a 3-kDa filter paper. The filtered samples were analysed for released silver ions using AAS (Perkin Elmer) [47].

2.9. Internalization of TiO_2 and Ag NPs in *Dunaliella* sp

Internalized Ag and TiO_2 NPs into *D. salina* were examined after 72 h of interaction with nanocomposite films under dark and UV-C condition. Post-interaction, the treated and control algal samples were centrifuged for 10 min at 7000 rpm , and the pellets were washed once with ASW to remove unbound NPs on the cell surface. Algal cell suspension was then acid digested with concentrated nitric acid and stored at $4\text{ }^\circ\text{C}$ for analysis of TiO_2 and Ag. The bioavailability of Ti^{4+} and Ag^+ was measured using atomic absorption spectroscopy (AAS) (Analyst 400, PerkinElmer) at a detecting wavelength of 325 and 334.94 nm for Ag and TiO_2 , respectively.

2.10. Characterization of nanocomposite and pristine films

The chitosan nanocomposite films incorporated with a higher concentration of TiO_2 in the presence of constant Ag were analysed using XRD (PAN analytical Xpert Pro MRD diffractometer, Amsterdam, Netherlands), equipped with Cu radiation at a wavelength of 0.15406 nm . Post-interaction, the films were scanned throughout the 2θ range of $10\text{--}50^\circ$ at $0.5^\circ\text{ min}^{-1}$ scanning rate at room temperature. The surface functional groups of treated and untreated pure chitosan and chitosan nanocomposite (TiO_2/Ag) films were examined by attenuated total reflectance-Fourier transform infrared (ATR-FTIR, Shimadzu, Japan) spectroscopy in the recording range of $400\text{--}4000\text{ cm}^{-1}$. After 72 h of interaction under UV-C and dark conditions, the films were rinsed with Milli-Q water to remove unwanted moisture and air dried for further examinations. The surface morphology of algal cells and that of untreated, dark, and UV-C exposed films was figured out using scanning electron microscope (SEM, ZEISS, India). After interaction, treated films and algal cells were washed with Milli-Q to remove the unbound algal cells and other biological components. Then, the samples were air dried on a glass slide and then analysed by SEM at a magnification scale of $28,000$. Topographical images of treated and untreated films were analysed using atomic force microscopy (AFM, Nano surf, Easy Scan 2, Switzerland) operated under static force mode and examined by ContA1-G cantilever type. The EZ2-AFM head type probe was used for analysis, and the software version 3.5.0.38 was used to process the images of AFM.

2.11. Statistical interpretation

The test results were assessed and represented as mean $\pm$ SE. Two-way ANOVA followed by Bonferroni's post-test method was used to analyse the level of statistical significance of all the experimental values. The values were accepted to be significantly different at a probability value less than 0.05 ($p < 0.05$).

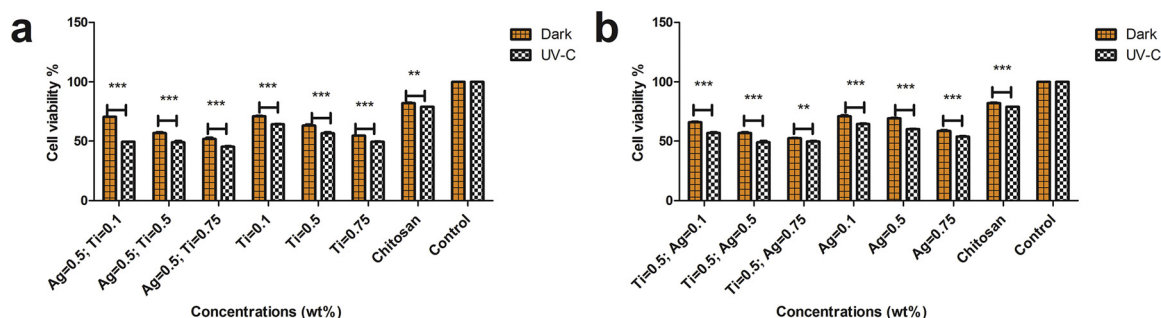


Fig. 1. Cell viability assessments (using MTT assay) showing a concentration-dependent decrease in the viability of *Dunaliella* sp. after treatment with chitosan nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films under dark and UV-C conditions. The values are mentioned as mean $\pm$ SE, and the asterisks symbols represent the level of significance ($p < 0.001 = ***$, $p < 0.01 = **$).

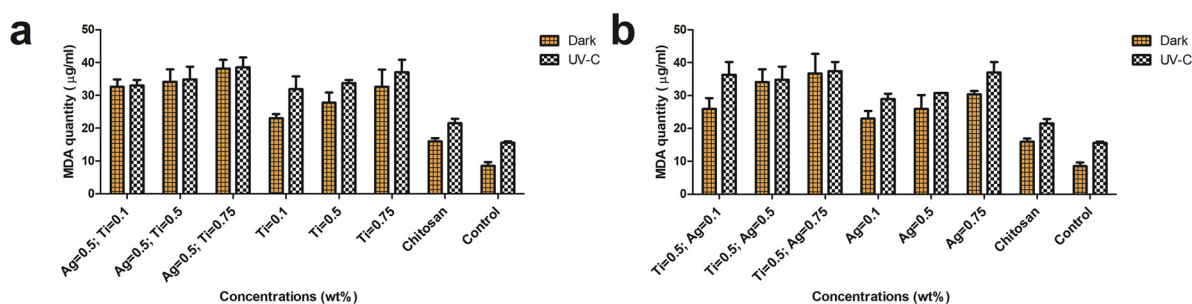


Fig. 2. Estimation of lipid peroxidation by quantifying MDA production in *Dunaliella* sp. treated with chitosan nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films under dark and UV-C conditions. The values are mentioned as mean $\pm$ SE.

3. Results and discussion

3.1. Anti-algal property of nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films

The anti-algal effects of pure chitosan, pristine (TiO₂ and Ag), and nanocomposite films displayed a decrease in viability of *Dunaliella salina* with respect to control under dark and UV-C condition (Fig. 1a and b). Under both the exposure conditions, loss of cell viability increased significantly ($p < 0.001$) with increasing concentration of pristine TiO₂ films. Under dark and UV-C exposure conditions, the highest concentration of TiO₂ pristine (0.75 wt%) showed cell viability of $54.75 \pm 0.09\%$ and $49.40 \pm 0.44\%$, respectively, when compared with the control (only algae) samples (100% viable cells). The nanocomposites with varying concentrations of TiO₂ and fixed concentration of Ag (0.1 and 0.75 wt% TiO₂/0.5 wt% Ag) showed significant decrease in ($p < 0.001$) cell viability under dark and UV-C, when compared with the reverse (0.1, and 0.75 wt% Ag/0.5 wt% TiO₂). Ag pristine film also showed a similar trend as that of TiO₂ pristine, i.e. concentration-dependent decrease in cell viability. However, when compared to TiO₂ pristine, Ag pristine showed significantly ($p < 0.001$) lesser decrease in cell viability for 0.5 and 0.75 wt% concentration.

In all the cases, nanocomposites (TiO₂/Ag) produced greater anti-algal effects on *D. salina* when compared with chitosan or pristine films (TiO₂ and Ag) alone. Another important observation was that UV-C exposure caused more death in algal cells when compared with dark condition ($p < 0.001$). Graziani et al. (2014) reported that high-intensity UV irradiation increases the photocatalytic activity of nano TiO₂-coated substrates and this finding correlates well with our observation, wherein significantly higher cell death was observed for TiO₂ pristine-treated algal cells rather than Ag pristine-treated cells under dark and UV-C conditions [54]. Moreover, UV radiation can induce the production of reactive oxygen species (ROS) in TiO₂ NP suspension, which in turn can cause more cell death [55]. From our study, it was also observed that TiO₂ NPs in the nanocomposite enhances the anti-algal effects of Ag under both exposure conditions. Earlier studies

proved that noble metals like Au or Ag help in achieving efficient charge separation by light-activated electron-hole pairs, which in turn enhance the photocatalytic effect of TiO₂ NPs [56–58]. The viability of *Dunaliella* sp. were decided by the factors of accumulation and/or internalization of nanoparticles from incorporated film, and quantity of cell membrane damages caused by nanoparticles and/or ions interaction with algal cells. The hydroxyl radicals produced by TiO₂ NPs affects the intra and extracellular contents such as enzymes, nucleic acids, phospholipids, peptidoglycans, and polysaccharides, which are reacting with polyunsaturated fatty acids inducing peroxidation and resulting in strong antimicrobial effects [59]. The leaching of Ag⁺ ions from Ag NPs, incorporated chitosan pristine and nanocomposite films influences the viability of algal cells.

3.2. Cell membrane damage

The damage of cell membranes by nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films was measured indirectly by estimating the MDA released under dark and UV-C condition in *Dunaliella* sp. (Fig. 2a and b). It was observed that under both the exposure conditions, the increase in TiO₂ composite concentrations (0.5 wt % Ag; 0.1, 0.5, 0.75 wt % TiO₂) increased the MDA released ($p < 0.001$). Higher concentration of TiO₂ NPs in the nanocomposite (0.5 wt% Ag/0.75 wt % TiO₂) showed MDA release values of $38.18 \pm 2.5 \mu\text{g/mL}$ and $38.55 \pm 2.9 \mu\text{g/mL}$ under dark and UV-C conditions, respectively. Similarly, for dark and UV-C exposure, increased concentration of Ag in composite (0.5 wt % TiO₂; 0.1, 0.5, 0.75 wt % Ag) produced higher MDA ($p < 0.001$). The obtained MDA values for higher Ag concentration (0.5 wt % TiO₂; 0.75 wt % Ag) in nanocomposite were $36.70 \pm 5.9 \mu\text{g/mL}$ and $37.44 \pm 2.7 \mu\text{g/mL}$ under dark and UV-C conditions, respectively. A similar concentration-dependent increase in MDA production was observed for both pristine (TiO₂ and Ag) films under dark and UV-C exposure condition ($p < 0.001$). The pure chitosan, pristine (TiO₂ and Ag), and nanocomposite (TiO₂/Ag) films for each concentration individually did not show any significant difference in MDA production under dark and UV-C exposure condition

($p > 0.05$). Similar to our study, MDA contents in the TiO_2/Ag NF photocatalytic reaction system was 2.0 times greater upon interaction with the algal cells after irradiation ultimately leading to oxidative stress and lipid peroxidation [60]. Oukarroum et al. found 45% of cell membrane damage to *D. tertiolecta* sp., upon interaction with Ag NPs at 1 mg/L concentration [61]. When both TiO_2 and Ag NPs are present in the films, ROS production was induced inside the cells [61,62]. ROS induction occurs by various factors such as higher salt concentration, higher light intensity, extreme temperatures, UV irradiation, heavy metal and NPs toxicity etc. [63]. This ROS in the biological systems eventually induced the cell membrane damages [64]. The mechanism of cell damage is that the oxidative radicals interact with cell membrane thereby increasing their permeability and causing leakage of small molecules, and ions from the cell. Eventually the degraded cellular components leads to cell death [65]. Accumulation of nanoparticles in the cell membrane provokes toxicity by cell wall damage and hetero-agglomeration [66–68]. Algae secretes EPS which envelopes the cells and help in the NP-cell agglomeration by linking the algal cells and NPs via adsorption on the NP surface. Stronger the hetero-agglomeration between algae and NPs, greater is the toxicity [69].

3.3. Quantification of extracellular polysaccharides (EPS)

Fig. 3a and b depicts the EPS production by the algal cells after 72-h interaction under dark and UV-C conditions with chitosan, pristine (TiO_2 and Ag), and nanocomposite (TiO_2/Ag) films. In all the cases, algal culture exposed under UV-C produced a higher amount of EPS (not significant for all conditions) when compared with that kept under dark condition. Pristine and nanocomposite films, after interaction with algal culture under UV-C, significantly enhanced ($p < 0.01$) the EPS production when compared with the control algal sample. EPS production was higher for UV-C exposed samples when compared to samples in dark condition. EPS production was also increased with increase in the concentration of NPs in pristine or nanocomposite films. For all concentrations of TiO_2 pristine films, EPS level was high (non-significant; $p > 0.05$) when compared with Ag pristine films under both dark and UV-C conditions, except for 0.1 wt% of TiO_2 pristine, which showed a significant increase in EPS under UV-C exposure when compared with Ag pristine films ($p < 0.05$). EPS production for nanocomposites (0.75 wt% $\text{TiO}_2/0.5$ wt% Ag) incubated under dark and UV-C was found to be 1.78 ± 0.72 and $3.48 \pm 0.25 \mu\text{g mL}^{-1}$, respectively, which was higher (non-significant; $p > 0.05$) than nanocomposite (0.75% wt% Ag/0.5 wt% TiO_2) films.

EPS was secreted by the algal cell and thickens up the cell membrane, which in turn acts as a protective mechanism towards nanoparticle-induced stress [70]. Though *D. salina* lacks a cell wall, EPS produced by the cells helps them to fight against the stress and did not make them vulnerable to contaminants [71]. In our study, we observed that EPS production increased with increase in concentration of pristine or nanocomposite films, which actually correlated well with previous findings, wherein they showed that upon exposure to contaminants,

EPS adsorbs them and thus prevent internalization of those contaminants [72,73]. Other factors also play a role in the production of EPS. For example Domozych, 2007 and Shen, 2015 evidenced that the increased concentrations of nutrient, especially nitrogen, induced more EPS production in green algae and diatoms species. The higher concentration of nitrogen in the medium induces the over production of amino acids because of the EPS containing proteins [74,75]. Additionally temperature, stress and rich minerals (calcium) also affects the production of EPS [74].

3.4. Assessment of antifouling activity

Biofouling, a complex process, involves the initial growth of microbial slime (biofilms) on the exposed wet surface of the submerged structures, which is then followed by the further attachment of algae and other different invertebrates [76]. The strong adhesion on membrane surface produces extracellular polymeric substances (EPS) made up of proteins, lipoproteins, glycoproteins, polysaccharides, and other biological macromolecules [77]. Multiplication of microorganism and production of EPS could lead to the biofilm formations [78–80]. To protect these submerged structures, it is important to breakdown the microbial biofilms. The antifouling property of chitosan, pristine (TiO_2 and Ag), and nanocomposite (TiO_2/Ag) films can be estimated by measuring the slime (loosely bound glycocalyx) formation and by calculating the mass of the films.

3.4.1. Estimation of slime formation

For all the treatments, we observed that UV-C-exposed samples inhibit the slime production significantly when compared with the same exposed under dark ($p < 0.001$) (Fig. 4a and b). Chitosan film-treated samples showed higher slime formation (0.98 ± 0.05 and 0.74 ± 0.02 under dark and UV-C) when compared with pristine- and nanocomposite-treated samples. Moreover, the slime formation reduced with the increase in the concentration of TiO_2 and Ag in pristine films. But interestingly under all exposure concentrations there was no significant ($p > 0.05$) reduction of glycocalyx production between pristine TiO_2 and pristine Ag as well as for nanocomposites (0.5 wt% $\text{TiO}_2/0.1$, 0.5, and 0.75 wt% Ag vs. 0.5 wt% Ag/0.1, 0.5, and 0.75 wt% TiO_2). Nanocomposites (0.5 wt% Ag/0.1 and 0.75 TiO_2) showed higher reduction in slime generation under dark condition ($p < 0.05$) when compared with pristine TiO_2 of similar concentration (0.1 and 0.75 wt%).

This increase in antifouling property of TiO_2/Ag nanocomposite can be supported by a previous study done by Graziani et al. (2016), wherein a colour variation method was used to prove the higher antifouling potency of TiO_2 -Ag specimen when compared with TiO_2 -Cu specimen [54]. Under UV-C exposure, there was no significant difference in inhibition of glycocalyx formation between pristine TiO_2 and pristine Ag films, though it was already proved that UV-C enhanced the photocatalytic property of TiO_2 NPs [81]. Chitosan coating also helps in increasing the antifouling activity, as an earlier study revealed that

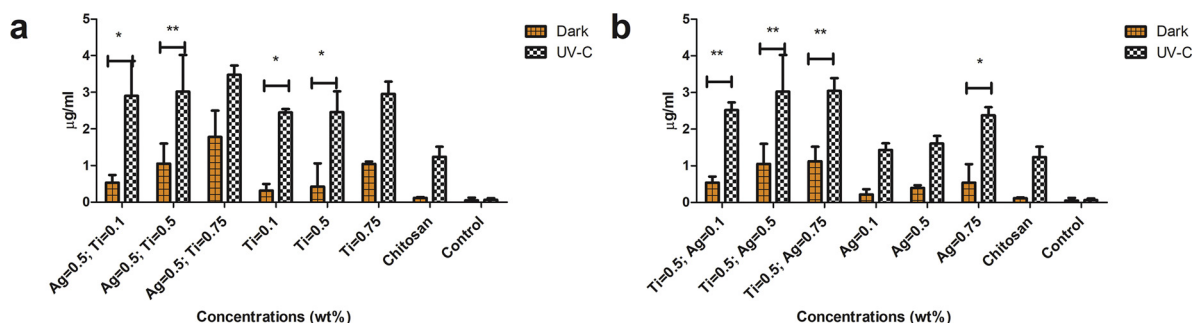


Fig. 3. EPS quantification in *Dunaliella* sp. treated with chitosan nanocomposite (TiO_2/Ag) and pristine (TiO_2 and Ag) films under dark and UV-C conditions. The values are mentioned as mean $\pm$ SE, and the asterisks symbols represent p values ($p < 0.01 = **$, $p < 0.05 = *$).

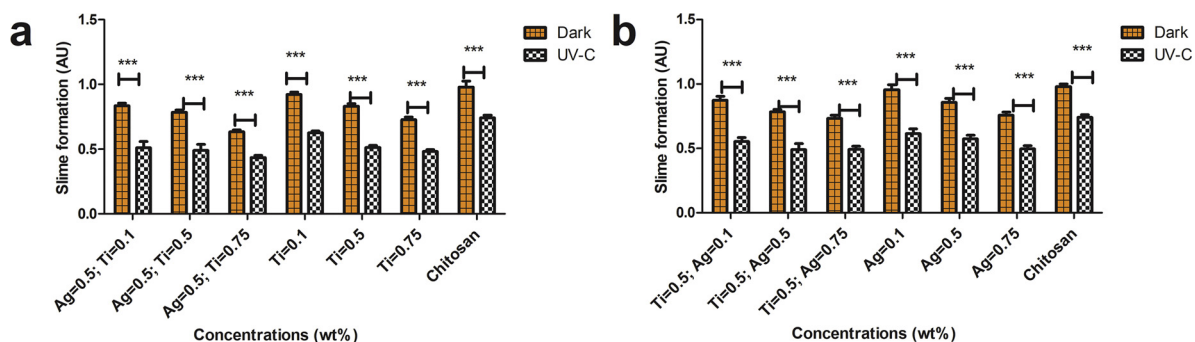


Fig. 4. Evaluation of glycocalyx (slime) formation by *Dunaliella* sp. treated with chitosan nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films under dark and UV-C conditions. The values are mentioned as mean $\pm$ SE, and the asterisks symbols represent p values ($p < 0.001 = ***$).

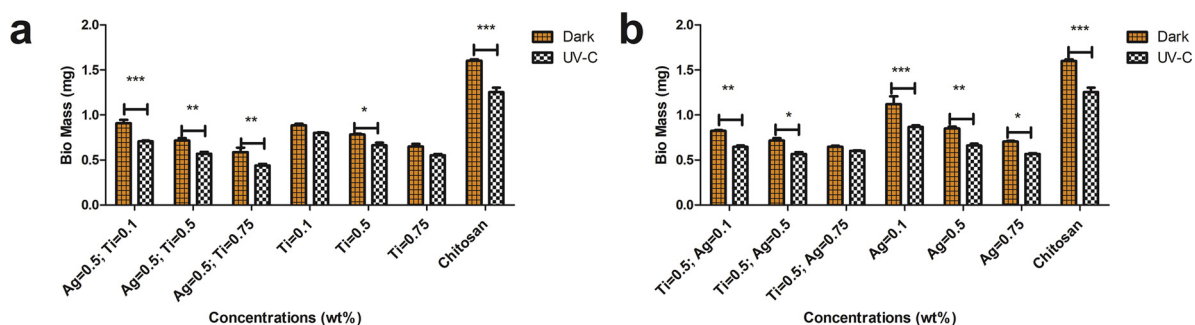


Fig. 5. Mass of adhered *Dunaliella* sp. on films after treatment with chitosan nanocomposite (TiO₂/Ag) and pristine (TiO₂ and Ag) films under dark and UV-C conditions. The values are mentioned as mean $\pm$ SE, and the asterisks symbols represent p values ($p < 0.001 = ***$, $p < 0.01 = **$, $p < 0.05 = *$).

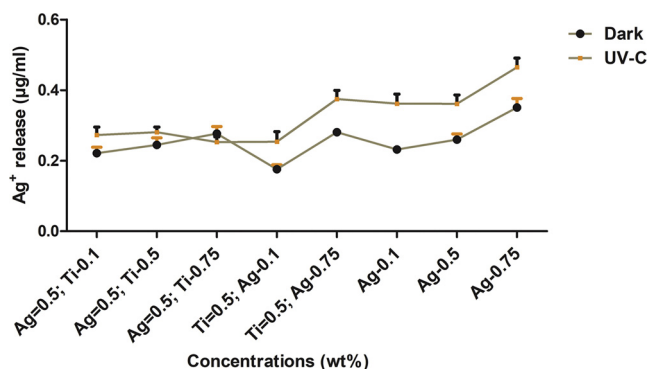


Fig. 6. Dissolution of Ag⁺ ions in aqueous medium from nanocomposite (TiO₂/Ag) and pristine (Ag) films after 72-h interaction in ASW under dark and UV-C conditions.

chitosan-coated Cu NPs inhibited algal growth (*Arthrospira*, *Chlorella*) by almost 90% [82]. Another study revealed that titanium nitride and silicon nitride when coated with Cu (TiSiN–Cu) inhibit the slime formation towards three different algal species (*N. closterium*, *P. triceratium*, and *Chlorella* sp.) by 93, 89, and 57%, respectively [83]. Ren et al. (2014) studied the fouling inhibition of Ag NPs coated structures against biofouling algal species of *Dunaliella tertiolecta* and the results showed 98.9% anti-adhesion activity by Ag NPs-PDA coated PS substrates [4]. The enhancement of antifouling activity by composite was also reported by Al-Naamani et al. They examined the anti-adhesion properties of chitosan-ZnO nanocomposite coatings in glass substrates against the adhesion of *Navicula* sp. (diatom) and the results indicates better anti-adhesion property of chitosan/ZnO nanocomposite compared to pure chitosan coatings [35]. These studies showed that anti-algal effect depends on both the algal species exposed as well as the NPs used.

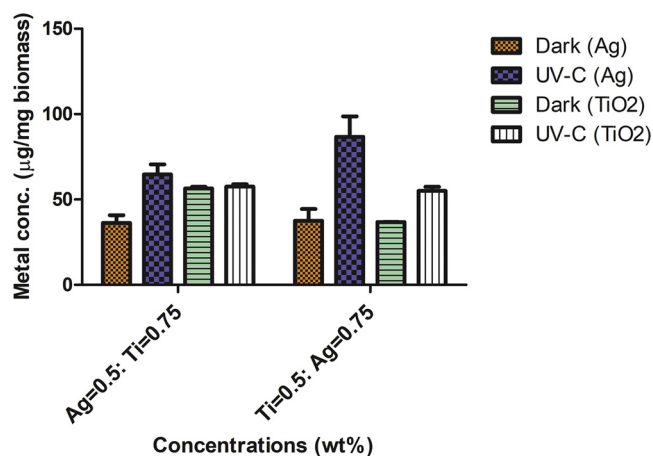


Fig. 7. Bioavailability of TiO₂ and Ag NPs in *Dunaliella* sp. treated with nanocomposite (TiO₂/Ag) films under dark and UV-C conditions. The values are mentioned as mean $\pm$ SE, and the asterisks symbols represent p values ($p < 0.01 = **$).

3.4.2. Mass assessment

After 72-h incubation under the dark and UV-C conditions, the mass of algal cells adhered to chitosan (blank), pristine (TiO₂ and Ag), and nanocomposite (TiO₂/Ag) films were estimated and represented in Fig. 5a and b. Pure chitosan (blank) showed algal masses of 1.6 ± 0.017 and 1.25 ± 0.049 under dark and UV-C. Under UV-C exposure, both pristine films (TiO₂ and Ag) showed less adhered algal biomass. Ag pristine showed significantly less biomass under UV-C for all three concentrations, whereas for TiO₂ pristine, one concentration (only 0.5 wt%) showed significantly less biomass under UV-C. Only for lower concentration (0.1 wt%) TiO₂ pristine showed less biomass when compared with Ag pristine ($p < 0.001$). Nanocomposite with higher concentrations of TiO₂ (0.5 wt% Ag/ 0.75 wt% TiO₂) showed

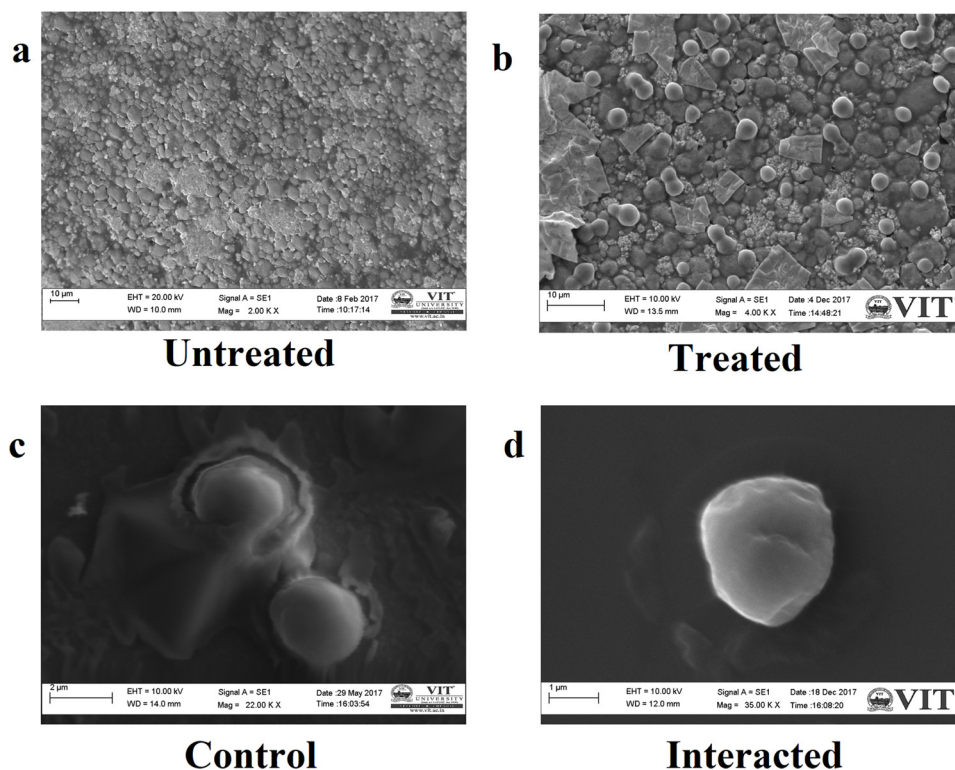


Fig. 8. Morphological analysis of *Dunaliella* sp. and nanocomposite (Ag = 0.5 wt%; TiO₂ = 0.75 wt%) film using SEM under dark and UV-C conditions: (a) untreated film, (b) treated film with algae, (c) control algae, (d) treated algae.

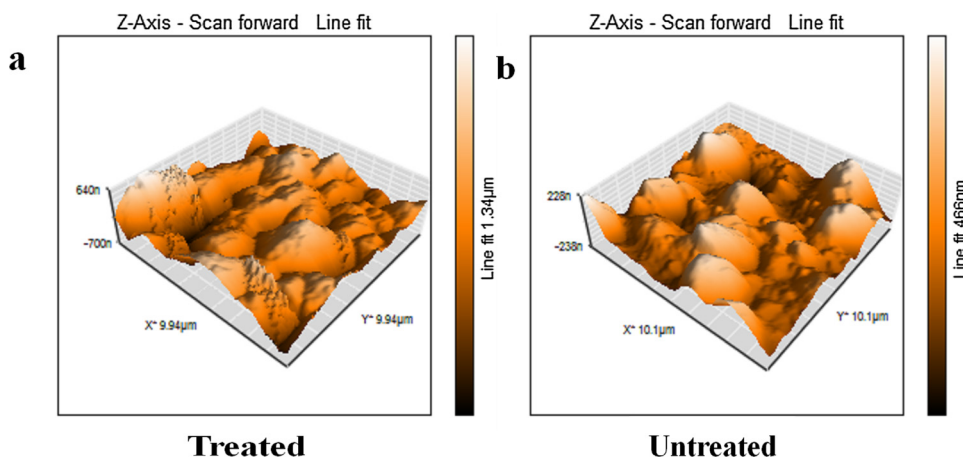


Fig. 9. Morphological analysis of nanocomposite (Ag = 0.5 wt%; TiO₂ = 0.75 wt%) films treated with *Dunaliella* sp. under UV-C condition using AFM: (a) treated and (b) untreated.

significantly less ($p < 0.001$) algal biomass under UV-C condition when compared with the opposite combination (0.5 wt% TiO₂/ 0.75 wt % Ag). It was also observed that with an increase in concentration of TiO₂ or Ag in the pristine film, the algal biomass adhered was less; however, not in all cases, the significant difference was observed. This observation revealed that nanocomposites with a higher amount of TiO₂ showed better antifouling activity. Similar report displayed a decrease in the mass of the MNP-doped films in comparison with their controls strongly indicating a reduced attachment of microbes over the film [53]. Physisorption of a furanone derivative over the polymer reduced the biofilm formation by *Staphylococcus epidermis* [84]. Thus the presence of TiO₂/Ag in the chitosan film enhances the antifouling activity by allowing lesser adhesion of algal cells on the nanocomposite.

3.5. Dissolution of silver ion (Ag⁺)

The release of the Ag ions from pristine (Ag) and nanocomposite (TiO₂/Ag) films in the medium was measured after the interaction time and is represented in Fig. 6. All combinations of nanocomposites, except the one with the highest concentration of Ag (0.5 wt% TiO₂ / 0.75 wt% Ag) showed non-significant release of Ag⁺ ion ($p > 0.05$) under dark and UV-C conditions. Only the concentration of 0.5 wt% TiO₂/ 0.75 wt% Ag showed a significant increase in the release of Ag⁺ ions under UV-C exposure when compared with dark condition ($p < 0.05$). For all the concentrations of pristine Ag films, the Ag⁺ release was significantly higher under UV-C in comparison with dark condition ($p < 0.001$ for 0.1 wt%, $p < 0.05$ for 0.5 wt%, and $p < 0.01$ for 0.75 wt% of AgNP concentration). The higher release observed under UV-C compared to dark condition may be attributed to

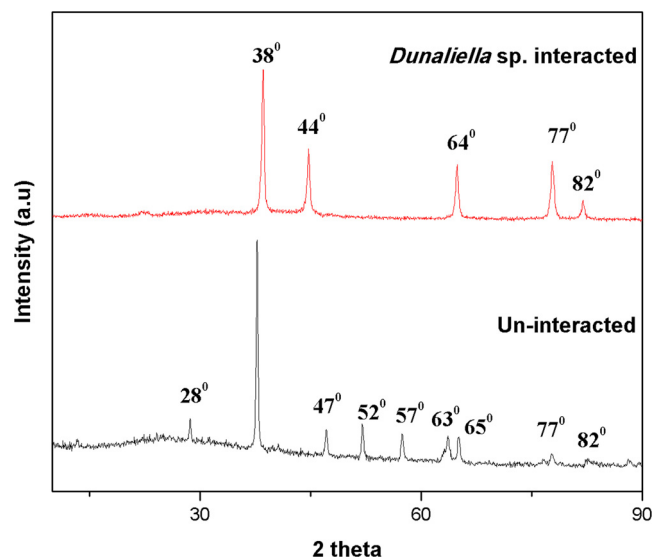


Fig. 10. XRD pattern of control and nanocomposite (Ag = 0.5 wt%; TiO₂ = 0.75 wt%) films treated with *Dunaliella* sp. under UV-C condition.

the photoactive nature of Ag⁺ ions. The photoactive property helps in the disinfection of viruses and bacteria [85]. The release of Ag⁺ ions from pristine Ag films were higher compared to the release from the nanocomposites. Similar observations were recorded by Egger et al. wherein the release of Ag⁺ ions from silver nitrate and silver zeolite in aqueous solution was faster compared to the release from Ag NPs embedded within the silica matrix [86]. Whereas in contrast to our study, the release of silver ions after 21 days from the Ag/TiO₂ embedded in MPTS and TEOS sol-gel matrix was in very low quantity [44]. The major mechanism of action of silver ions is that it interacts with the thiol groups of proteins and deactivates the respiratory enzymes and also prevents DNA replication [87,88].

3.6. Uptake of TiO₂ and Ag NPs inside algal cells

After 72-h interaction with chitosan nanocomposite films under dark and UV-C, the cellular internalization of TiO₂ and Ag by the *Dunaliella* sp. was noted (Fig. 7). The highest uptake of TiO₂ NPs was observed in algal cells after interaction with higher concentrations of TiO₂ (0.75 wt %) in the presence of constant Ag concentration (0.5 wt %) and was observed to be 56.48 ± 0.92 µg/mg and 57.61 ± 1.3 µg/mg under dark and UV-C conditions, respectively. In contrast, the lowest uptake of TiO₂ was observed upon interaction with higher concentration of Ag (0.75 wt %) in the presence of constant TiO₂ concentration (0.5 wt %) for nanocomposite films under dark (36.76 ± 0.23 µg/mg) and UV-C (55.11 ± 2.3 µg/mg) irradiation, respectively. The concentration-dependent uptake of TiO₂ by algal cells

was observed to be non-significant at both concentrations of nanocomposites (TiO₂/Ag and Ag/TiO₂) ($P > 0.05$) under both exposure conditions.

Similarly, the highest uptake of Ag was noted after interaction of *Dunaliella* sp. with nanocomposite (0.5 wt% TiO₂/ 0.75 wt% Ag) under UV-C (86.66 ± 12.01 µg/mg) and dark (37.61 ± 6.83 µg/mg) conditions, respectively. The lowest uptake of Ag in *Dunaliella* cells upon interaction with a higher concentration of TiO₂ (0.75 wt %) in the presence of constant Ag concentration (0.5 wt %) was found to be 36.25 ± 4.61 µg/mg and 64.66 ± 5.92 µg/mg under dark and UV-C conditions. The concentration-dependent non-significance was observed for both nanocomposites under dark and UV-C conditions ($P > 0.05$). The interaction condition-dependent significant difference was observed for both nanocomposite (TiO₂/Ag and Ag/TiO₂) films ($P < 0.001$). The accumulation of TiO₂ NPs inside the algae shows direct physical effects like cell wall damages or indirect effects such as reduction in the cell count and/or the restriction of metal uptake by microalgae [67,89,90]. Though TiO₂ NPs showed higher cytotoxicity and cell membrane damage, uptake of Ag was higher than TiO₂ under UV-C exposure conditions. This could happen because of the higher cell viability of Ag-treated cells, wherein during reproduction of daughter cells, some alteration in the membrane permeability allows the NPs to get attached to the cell wall [91].

3.7. Characterization of nanocomposites and pristine films

After interaction of *Dunaliella* sp. with the highest concentration of nanocomposite (0.5 wt% Ag/ 0.75 wt% TiO₂), the morphological changes were observed using SEM (Fig. 8). Fig. 8a depicts the untreated nanocomposite film surface with a uniform distribution of Ag and TiO₂ NPs, and Fig. 8b demonstrates the adhesion of algal cells to the films after interaction with the nanocomposite. In Fig. 8c, intact spherical shaped algal cells without any membrane damage were observed, whereas algal cells after treatment showed the adsorption of NPs on the adhered cells (Fig. 8d). Similar to our study, morphological analysis of silver nitrate treated PDA substrates showed the binding of Ag NPs on their surface [4].

The antifouling activity of TiO₂/Ag nanocomposite can be confirmed by comparing the morphology of the treated films with that of untreated one, i.e. less slime formation was observed. Surface roughness and morphology of *Dunaliella* sp. interacted with nanocomposite film containing a higher concentration of TiO₂ in the presence of a constant concentration of Ag (Ag = 0.5 wt%; Ti = 0.75 wt %) under UV-C condition were evaluated by AFM and compared with respect to untreated nanocomposite film (Fig. 9 a and b). The roughness of the untreated nanocomposite film (61.01 nm) was compared with treated nanocomposite films, which had increased roughness (143.4 nm). Wang et al. confirmed the single layer attachment of algal cells over the TiO₂/Ag nanofiber mats before and after the loading of algal cells. Algae decorated TiO₂/Ag hybrid nanofiber membrane with enhanced photocatalytic activity for Cr (VI) removal under visible light. Thus the

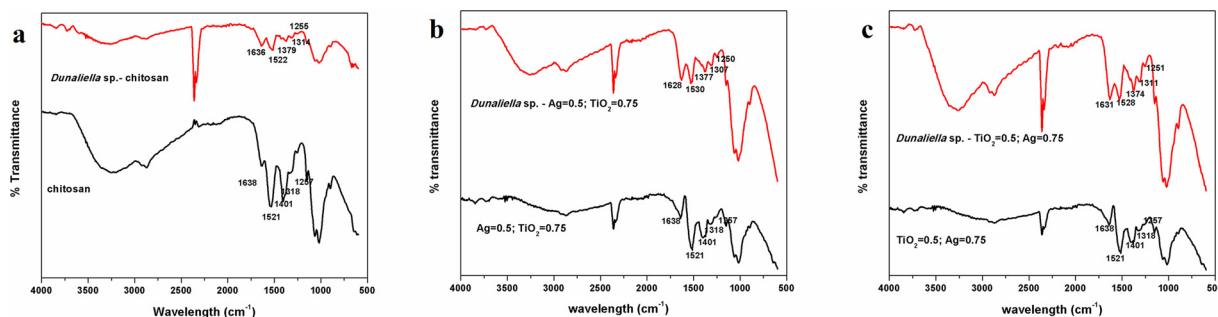


Fig. 11. FTIR spectra showing the functional group changes in *Dunaliella* sp. treated with (a) pure chitosan, (b) 0.5 wt% TiO₂/ 0.75 wt% Ag nanocomposite, and (c) 0.5 wt% Ag/ 0.75 wt% TiO₂ nanocomposite under UV-C condition.

topography of interacted and un-interacted nanocomposite films could explain how the fouling organism influenced the roughness of the substrate surface [60].

Fig. 10 indicates the XRD peaks of untreated nanocomposite (TiO_2/Ag) film and that of films after interaction with algae under UV-C condition. The graph showed changes in the peaks after treatment with algae when compared with peaks of untreated films, all of which suggested that algal interaction caused changes in the structures of the films. The common peak of 38° represents the plane for Ag (111) and the un-interacted film planes 105° (52°), 110° (57°), 204° (63°), and 220° (65°) indicated the face-centred cubic TiO_2 and Ag peaks [47].

FT-IR analysis showed changes in the functional groups for pure chitosan (blank) in comparison with chitosan films treated with *Dunaliella* sp. under UV-C exposure (Fig. 11a). Two specific peaks representing a monocarboxylic group (C–O) stretching band were noticed at 1021 and 1065 cm^{-1} for both algal treated and untreated nanocomposite films. Amide II stretching bands (C–N and C–O) were observed at the wavelength of 1324 and 1528 cm^{-1} for both algal treated and untreated films. The presence of trimesoyl ester groups (C–O) was suggested by the peaks at 1719 and 1918 cm^{-1} . The wavenumbers at 1631 and 1374 cm^{-1} represent the amide groups of C=O and C–O [92]. Monocarboxylic vibration band was represented by the peak at 2355 cm^{-1} . Two small peaks for the aliphatic C–H stretch band were noted at 2864 and 2925 cm^{-1} [93]. Two small peaks at 3722 cm^{-1} and 3856 cm^{-1} were designated for –OH stretching bands. All the represented amide bands are characteristic of chitosan [93]. Similar peaks were also observed for nanocomposites (0.5 wt% TiO_2 /0.75 wt% Ag and 0.5 wt % Ag/0.75 wt% TiO_2) (Fig. 11b and c).

4. Conclusion

The present study revealed that chitosan nanocomposite incorporated with TiO_2 and Ag NPs can be used as a potential antifouling agent against marine algae. The antifouling effects of the nanocomposites and pristine films can be perceived based on the decrease in cell viability, cell membrane damage, and prevention of algal biomass formation. Ag NPs help in enhancing the photocatalytic property of TiO_2 NPs, and hence, the nanocomposite showed better anti-algal property when compared with pure chitosan or pristine films used in this study. In most of the cases, the effect of TiO_2 NPs dominated over Ag NPs and showed more antifouling properties as UV-C irradiation can increase the photocatalytic effects of TiO_2 NPs.

Conflict of interest

The authors declare that they have no conflict of interest.

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Appendix A. Supplementary data

Supplementary material related to this article can be found, in the online version, at doi:<https://doi.org/10.1016/j.jece.2018.10.050>.

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